

Real-Time Weather App - Project Synopsis

Project Title: ■ Real-Time Weather App

Objective:

The objective of this project is to build a user-friendly, real-time weather application that allows users to

Tools and Technologies Used:

- Python: Programming language used for scripting and logic.
- Streamlit: Framework for building interactive web apps.
- Requests Library: For HTTP requests to the API.
- OpenWeatherMap API: Provides real-time weather data.
- Datetime Module: Used for formatting timestamps.

Key Features:

1. User Input:

Allows users to input a city name to retrieve weather details.

2. API Integration:

Connects to OpenWeatherMap using an API key to fetch data.

3. Data Processing:

Extracts temperature, humidity, wind speed, pressure, sunrise, sunset, and description.

4. Error Handling:

Provides user-friendly error messages for incorrect city names or network issues.

5. Responsive UI:

Displays data with proper formatting and clarity.

Challenges Addressed:

- API key security and validation.
- Handling network and data errors gracefully.
- Formatting timestamps and weather descriptions.

Conclusion:

The Real-Time Weather App is a practical project demonstrating API integration, error handling, and us